# EXPERIMENT 1

# SOFTWARE REQUIREMENT SPECIFICATION (SRS)

## Aim

To develop a **Software Requirement Specification (SRS) document for the project Stock Maintenance System**, which allows administrators and staff to manage product inventory, update stock records, and monitor stock availability efficiently.

# Objective

- To understand the system requirements of the stock maintenance system.
- To document functional and non-functional requirements.
- To analyze the system architecture.
- To define user interactions with the system.

# Tools Used

The following CASE tools can be used for system analysis and documentation:

- Draw.io
- Lucidchart
- StarUML
- Creately
- EdrawMax
- Microsoft Word

# Theory

A **Software Requirement Specification (SRS)** is a document that describes the behavior, functions, and constraints of a software system. It acts as a contract between developers and stakeholders.

The SRS document includes:

• System overview
• Functional requirements
• Non-functional requirements
• External interfaces
• System constraints

For the **Stock Maintenance System**, the SRS defines how administrators and staff interact with the system to add products, update stock, monitor inventory levels, and generate reports.

# System Description

The **Stock Maintenance System** is a web-based inventory management platform that enables users to:

• Add and manage product records
• Monitor stock levels
• Update product quantities
• Generate inventory reports
• Manage product data efficiently

The system uses a **database server** to store product information and stock records.

# Overall Description

## Product Perspective

The Stock Maintenance System follows a **client-server architecture**:

User Interface → Application Server → Database

The application server processes stock operations while the database stores product and inventory information.

# Product Functions

The system provides the following functions:

• User registration and login
• Add new products
• Update stock quantity
• View product inventory
• Generate inventory reports
• Admin management

# User Classes

| User Type | Description |
|---|---|
| Staff | Manages product stock and updates inventory |
| Admin | Controls system data and generates reports |

# Functional Requirements

- **User Registration:** Users must be able to create an account in the system.
- **Login Authentication:** Users must log in using username and password.
- **Add Product:** Admin or staff can add new product details.
- **Update Stock:** The system must allow updating stock quantities.
- **Inventory Monitoring:** Users can view product stock information.
- **Report Generation:** Admin can generate inventory reports.

# Non-Functional Requirements

- **Performance:** The system should update stock records quickly.
- **Security:** User information must be protected using secure authentication.
- **Reliability:** The system must maintain accurate inventory records.
- **Usability:** The system interface should be simple and easy to use.

# System Constraints

- Internet connectivity required
- Database server required
- Browser compatibility required

# EXPERIMENT 2

# DATA FLOW DIAGRAM (DFD)

## Aim

To develop **Data Flow Diagrams (DFD Level 0 and Level 1)** for the Stock Maintenance System.

# Objective

- To understand system data flow
- To identify processes and data stores
- To represent system interactions visually

# Tools Used

- Draw.io
- Lucidchart
- Creately
- EdrawMax

# Theory

A **Data Flow Diagram (DFD)** shows how data moves through a system.

It represents:

• Processes
• Data stores
• External entities
• Data flow

DFDs help understand the logical structure of the system.

# Level 0 DFD (Context Diagram)

The **Level 0 DFD** represents the entire system as a single process interacting with external entities.

## External Entities

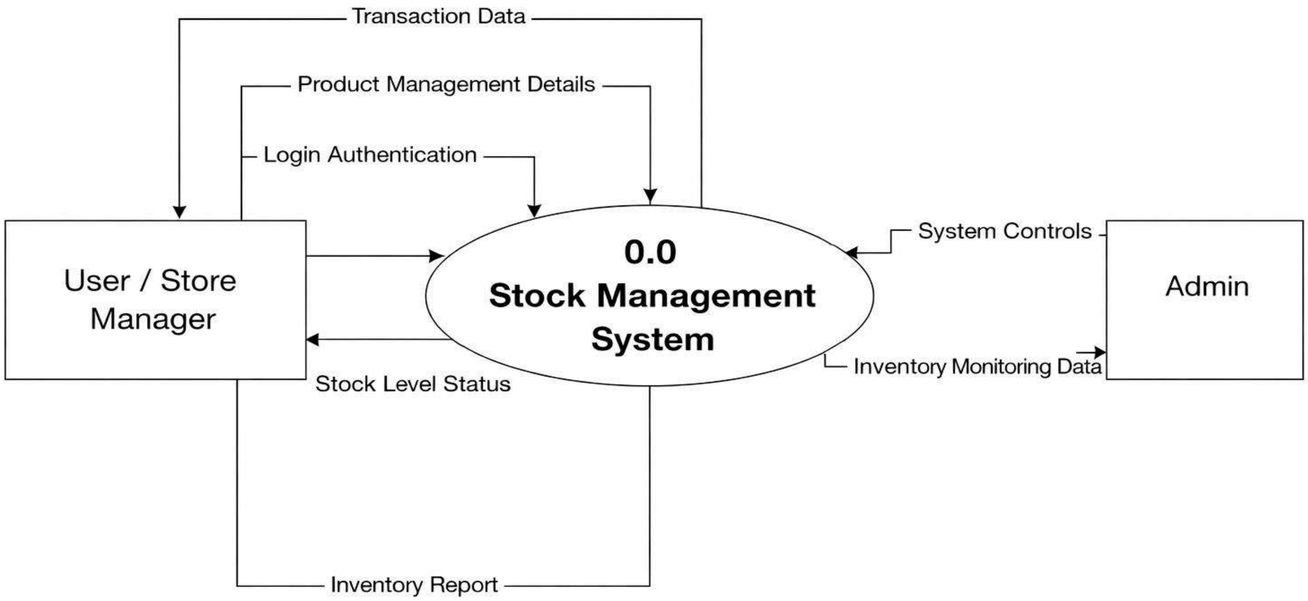
• User / Staff
• Admin

## Main Process

• Stock Maintenance System

### Explanation

The Level 0 DFD shows how users interact with the stock maintenance system to update inventory and manage product records while administrators monitor system operations.

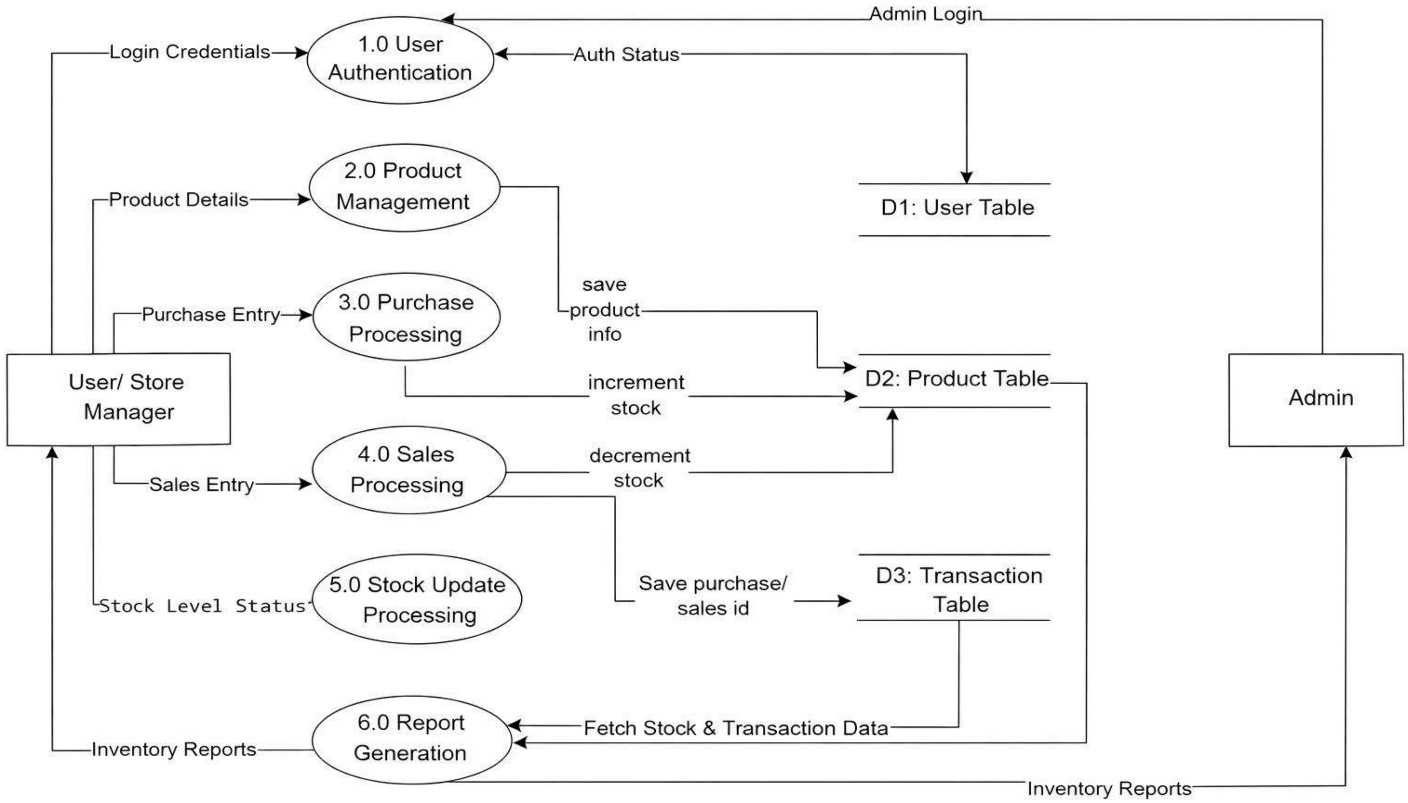



## Level 1 DFD

The **Level 1 DFD** breaks the system into major processes.

Processes include:

- User Authentication
- Product Management
- Stock Update Processing
- Inventory Monitoring
- Report Generation



Admin Login
Login Credentials
1.0 User Authentication
Auth Status
2.0 Product Management
Product Details
D1: User Table
save product info
Purchase Entry
3.0 Purchase Processing
User/ Store Manager
increment stock
D2: Product Table
Admin
4.0 Sales Processing
Sales Entry
decrement stock
Save purchase/ sales id
D3: Transaction Table
Stock Level Status
5.0 Stock Update Processing
6.0 Report Generation
Fetch Stock & Transaction Data
Inventory Reports
Inventory Reports

# Data Dictionary

| Data Element | Description |
|---|---|
| ProductID | Unique identifier for product |
| ProductName | Name of the product |
| Quantity | Number of available items |
| Supplier | Product supplier details |
| StockStatus | Current stock availability |
| ReportData | Inventory report information |

# Result

The **Data Flow Diagram model for the Stock Maintenance System was successfully developed.**

# EXPERIMENT 3

# STRUCTURED DESIGN

## Aim

To develop the **structured design model for the Stock Maintenance System based on the DFD**.

# Objective

- To convert DFD into system modules
- To design hierarchical module structure

# Tools Used

- Draw.io
- Creately
- EdrawMax
- Lucid chart

# Theory

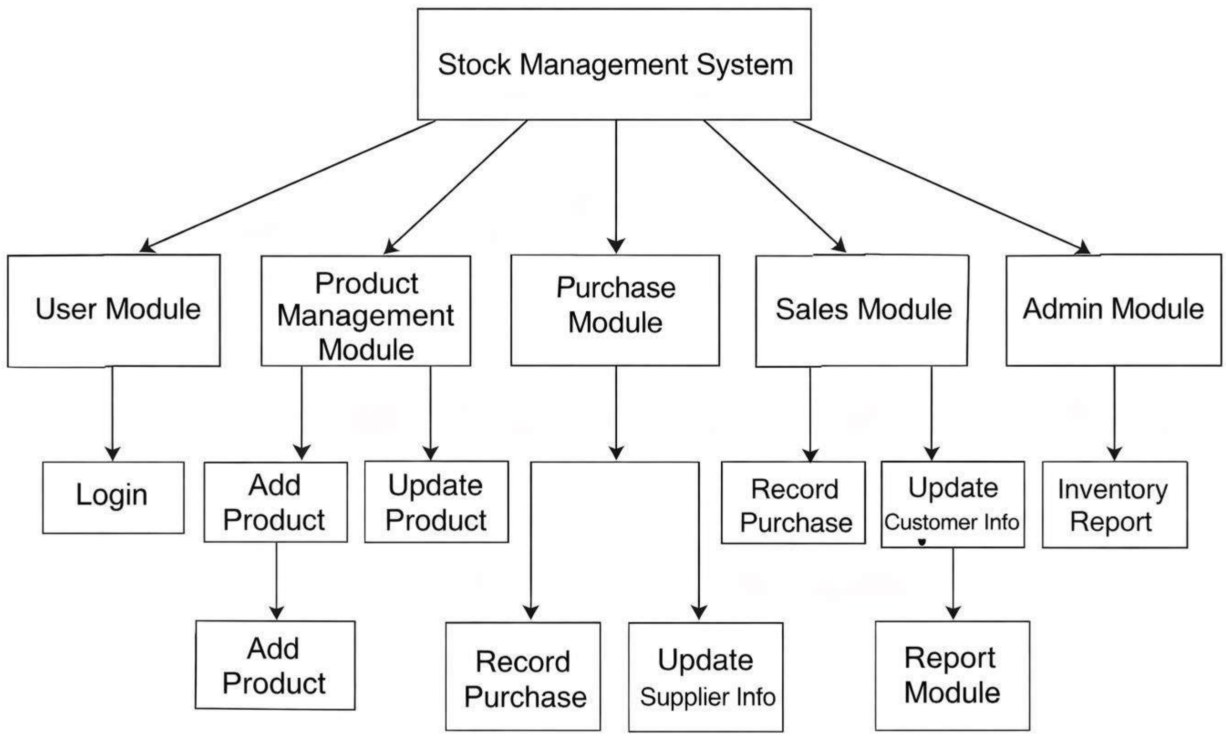
Structured design divides the system into hierarchical modules to simplify development.

Each module performs a specific function in the system.

# Structure Chart

The **Stock Maintenance System consists of the following modules:**

- User Module
- Product Module
- Stock Management Module
- Inventory Monitoring Module
- Report Module
- Admin Module




## Module Description

| Module | Description |
|---|---|
| **User Module** | Handles user authentication |
| **Product Module** | Manages product records |
| **Stock Management Module** | Updates product stock |
| **Inventory Monitoring Module** | Tracks stock levels |
| **Report Module** | Generates stock reports |
| **Admin Module** | Controls system operations |

## Result

The structured design of the **Stock Maintenance System was successfully developed.**

# EXPERIMENT 4

# UML USE CASE MODEL

## Aim

To develop a **UML Use Case Model for the Stock Maintenance System**.

# Objective

- To identify system actors
- To define system interactions
- To represent system functionality

# Tools Used

- StarUML
- Draw.io
- Lucidchart
- Creately

# Theory

A Use Case Diagram typically includes:

- Actors – The users or external systems interacting with the application.

- Use Cases – The actions performed by the actors.

- Relationships– The connections between actors and use cases.

It helps in understanding system functionality from the user perspective.

# Actors

| Actor | Description |
|---|---|
| **User / Store Manager** | Manages products , stock , purchases , and sales |
| **Admin** | Manages system data and monitors inventory operations |

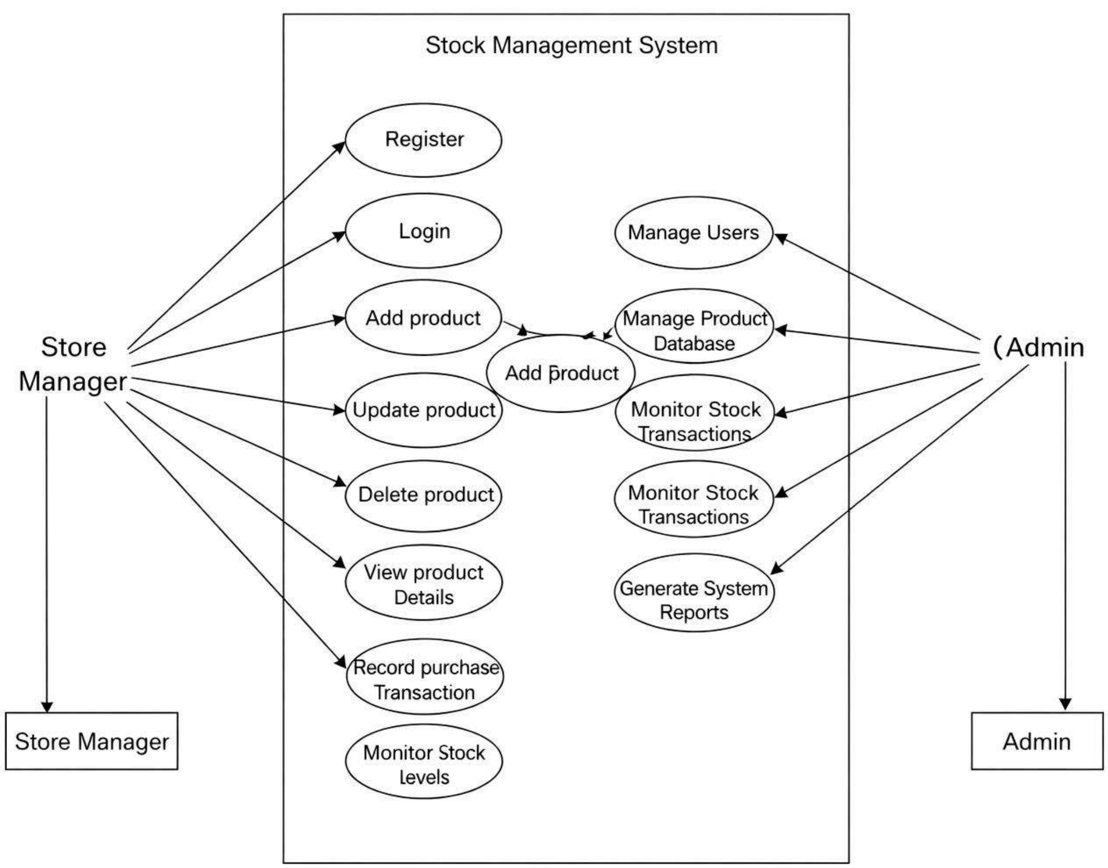
# Use Cases

## User / Store manager can:

1. Register
2. Login
3. Add Product
4. Update Product
5. Delete product
6. View product details
7. Record purchase transaction 
8. Record sales transaction
9. Monitor stock levels
10. Generate inventory reports

## Admin can:

1. Manage users
2. Manage product data base
3. Monitor stock transactions
4. Generate system reports



# Explanation

The Use Case Diagram illustrates how users interact with the Stock Management System to manage products , update stock , record purchases and sales , and generate inventory reports. The administrator manages system data and over sees overall inventory operations.

## Result

The UML Use Case Model for the Stock Management System was successfully developed.